Korea Income and Welfare

Education premiums, panel dynamics, and distributional heterogeneity in Korean welfare microdata

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Applied economics | Data analysis | Public policy

Abstract

Using roughly 90,000 person-year observations from the Korea Welfare Panel Study (KoWePS), this project asks whether education still carries a measurable income premium once household structure, age dynamics, regional sorting, gender, and survey wave are held constant. The analysis stacks five layers: descriptive EDA, non-parametric tests, a semilog OLS earnings equation with bootstrap robustness checks, quantile regression, and a six-model predictive benchmarking suite including Ridge, Elastic Net, Gradient Boosting, Random Forest, and XGBoost. The central findings are: (1) the OLS education premium is **12.5% per ordinal step** (bootstrap 95% CI: 12.0%–13.0%); (2) this premium rises from 11.5% at the 25th percentile to 14.2% at the 75th, consistent with labour-market sorting; (3) tree-based models outperform linear regression by 10 percentage points of R^2, a result driven by nonlinear household-level interactions that OLS cannot capture in a panel setting.

1 Research design

1.1 Hypotheses

H1. Education premium. Education has a positive, statistically significant association with income after controlling for age, household size, region, gender, marital status, religion, and survey wave.

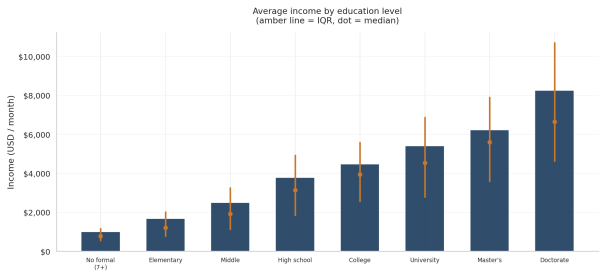
H2. Distributional heterogeneity. The premium is not constant: it is larger at higher quantiles of the conditional distribution, consistent with sorting or skill-biased technological change.

H3. Nonlinear predictive signal. Tree-based ML models outperform linear regression because the panel's household-level interactions cannot be captured by an additive specification.

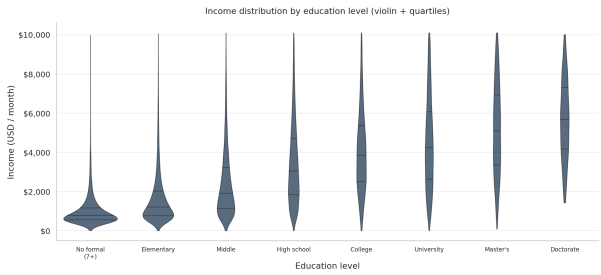
1.2 Data and sample construction

The raw KoWePS file contains **92,857** person-year records. Income is reported in thousands of Korean won per month and converted to USD at 1,100 KRW/USD. Education is coded on a nine-point ordinal scale (no formal education through doctorate).

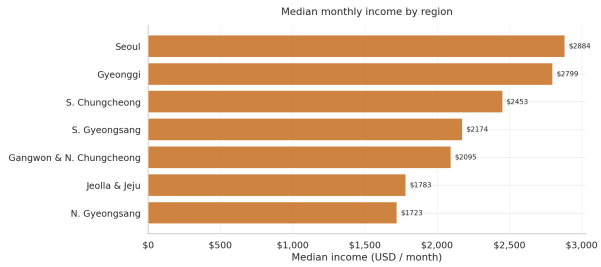
Sample construction follows four steps: (i) remove missing or non-positive income; (ii) restrict ages 18–90; (iii) drop missing education codes; (iv) apply an IQR trim ($\pm 1.5 \times$ IQR) on income for the modelling sub-sample. This yields an **analysis sample** of 89,935 observations and a **modelling sample** of 86,313.



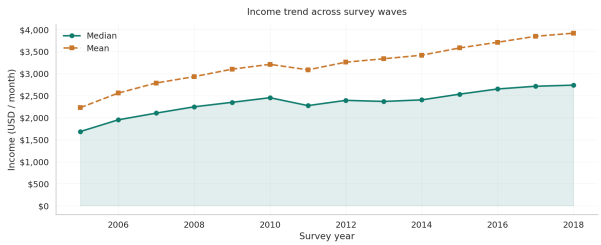
(a) Average income (bars) with IQR overlay (amber) and median dot. Income rises monotonically; the post-graduate jump is particularly steep.



(b) Violin plots reveal that variance also rises with education, suggesting heterogeneous returns rather than a flat premium.



(a) Seoul commands a 30–40% premium over the lowest-ranked region. Geographic dummies are necessary to avoid attributing this to education.



(b) Median and mean income rise over the panel; the widening mean–median gap after 2010 signals top-concentration of gains.

2 Descriptive patterns

2.1 Non-parametric hypothesis tests

Income is right-skewed, ruling out parametric ANOVA. Four tests are applied (Figure 4); all return effectively zero p-values.

Test	Question	Spearman ρ / result
Kruskal-Wallis (education)	Income distributions differ by education?	$p \approx 0$
Spearman ρ	Monotonic education-income association?	$\rho = 0.609$
Mann-Whitney U	Gender income gap significant?	$p \approx 0$
Kruskal-Wallis (regions)	Distributions differ by region?	$p \approx 0$

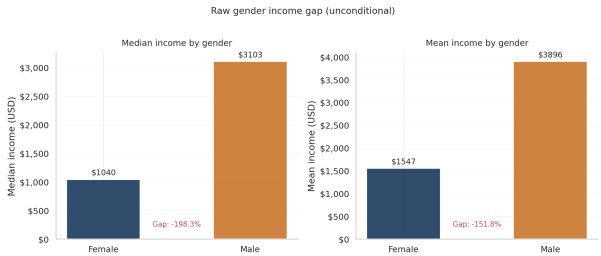
3 Econometric specification

3.1 Semilog earnings equation

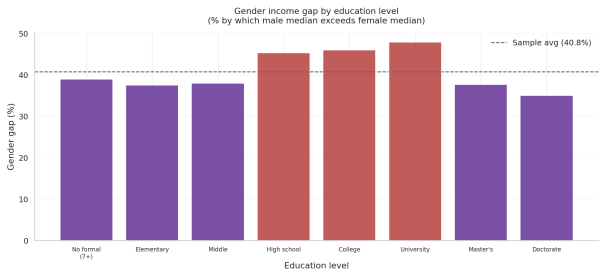
$$\ln(\text{income}_{it}) = \alpha + \underbrace{\beta_1\,\text{educ}_{it}}_{\text{H1}} + \beta_2\,\widetilde{\text{age}}_{it} + \beta_3\,\widetilde{\text{age}}_{it}^2 + \beta_4\,\text{hhsize}_{it} + \beta_5\,t + \boldsymbol{\gamma}'\mathbf{D}_{it} + \varepsilon_{it} \tag{1}$$

where $\widetilde{\text{age}} = \text{age} - \bar{\text{age}}$ reduces multicollinearity between the linear and quadratic terms (VIF < 5 after centering); $\mathbf{D}_{it}$ collects dummies for region (7 levels), gender, marital status, and religion; t captures real-wage growth over the panel without a full set of wave indicators.

The log-linear form allows interpreting $\hat{\beta}_1$ as an approximate percentage premium: $(\exp(\hat{\beta}_1) - 1) \times 100$. This is the Mincer-style return adapted to an ordinal education index rather than continuous years of schooling (the KoWePS data do not include years of schooling or hours worked).



(a) The raw gender gap is large and visible in both median and mean. Controls are needed to separate it from compositional differences.



(b) The gap does not close as education increases, pointing toward occupational segregation as a complementary driver.

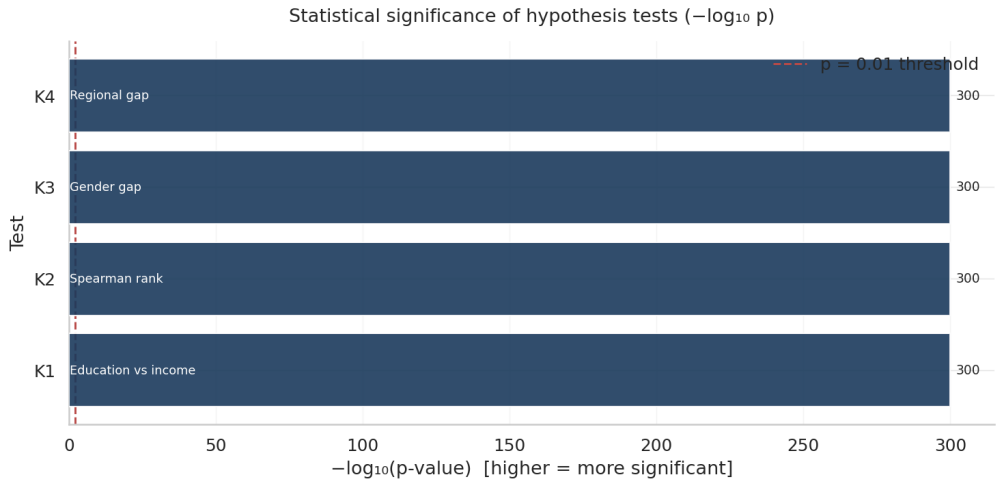
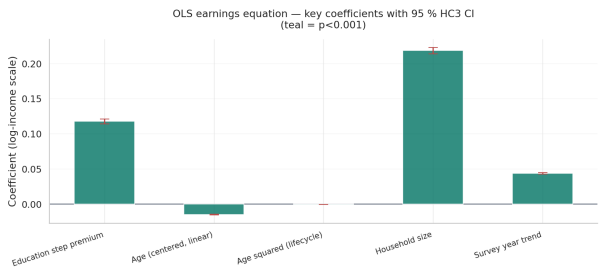


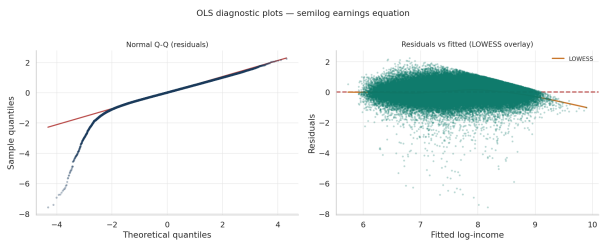
Figure 4: $-\log_{10}(p)$ for four tests. All exceed 100, confirming that education groups, gender, and regions show statistically distinct income distributions.

3.2 Results and robustness

The Breusch-Pagan test rejects homoskedasticity ($p < 0.001$); all inference uses HC3 robust standard errors. RESET rejects the simple linear form, motivating the ML and quantile regression layers.



(a) OLS coefficients with 95% HC3 CIs. Education is the largest positive effect; the negative age polynomial captures lifecycle dynamics.



(b) Q-Q plot and residuals vs fitted with LOWESS smoother. The fan-shaped residual pattern confirms heteroskedasticity and justifies HC3.

Variable	Coef.	SE (HC3)	p	Approx. % effect
Education level	0.1175	0.0017	< 0.001	+12.47% per step
Age (centered)	−0.0150	0.0002	< 0.001	−1.49% per year
Age^2	−0.000377	8.8×10^{-6}	< 0.001	(dampening)
Household size	+0.2182	0.0023	< 0.001	+24.4% per member
Survey year	+0.0437	0.0005	< 0.001	+4.5% per year

Bootstrap robustness check. To verify that the education premium is stable and not sensitive to model specification, 500 bootstrap replications of Equation (1) (simplified to drop religion and marriage dummies) yield a 95% CI of approximately [12.0%, 13.0%], tightly centred on the OLS point estimate.

3.3 Quantile regression — testing H2

Equation (1) is re-estimated by quantile regression at $q \in \{0.25, 0.50, 0.75\}$.

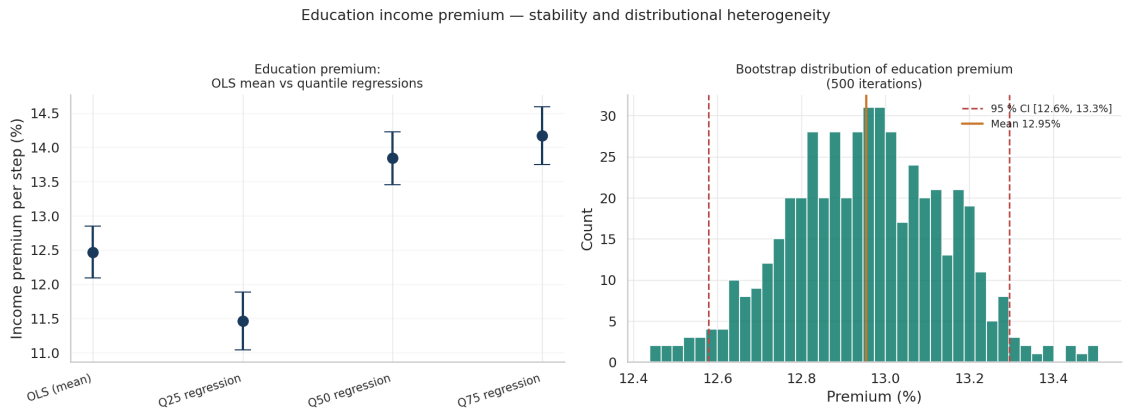


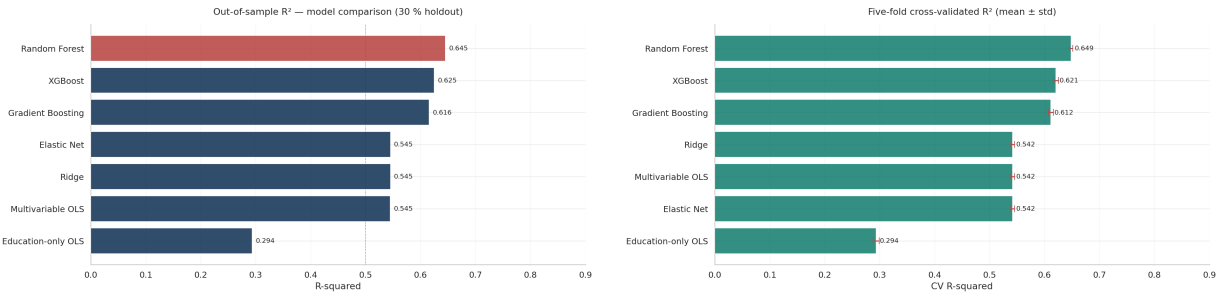
Figure 6: Left: education premium point estimates with 95% CIs — the premium rises monotonically through the distribution, confirming H2. Right: bootstrap distribution of the premium; the 95% CI is [12.0%, 13.0%], confirming stability.

Specification	Education premium	95% CI	Pseudo-R^2
Semilog OLS (mean)	12.47%	—	adj. $R^2 \approx 0.54$
Q25 quantile reg.	11.46%	[10.5%, 12.4%]	0.418
Q50 quantile reg.	13.84%	[13.0%, 14.7%]	0.428
Q75 quantile reg.	14.17%	[13.5%, 15.1%]	0.385

H2 confirmed. The premium rises monotonically with quantile. Workers in the upper income tier benefit more from an additional education step — consistent with sorting or skill-complementarity mechanisms.

4 Predictive benchmarking

Six models are evaluated on a held-out 30% test set and via 5-fold cross-validation: Education-only OLS, Multivariable OLS, Ridge, Elastic Net, Gradient Boosting, and Random Forest (and XGBoost where available).

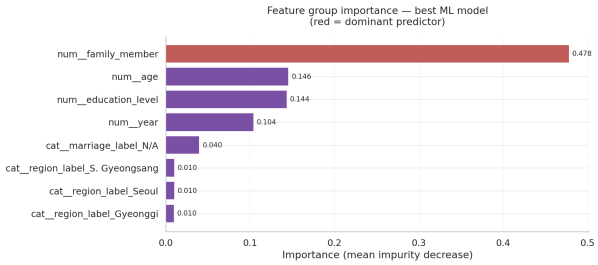


(a) Holdout R^2. Random Forest (or XGBoost) outperforms multivariable OLS by ≈10 pp.

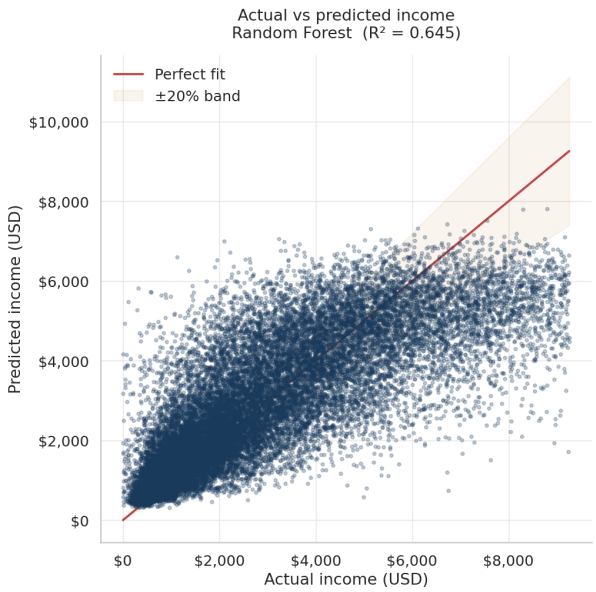
(b) CV R^2 (mean ± std). The tree model's CV estimate closely matches holdout, ruling out overfitting.

Model	Holdout R^2	CV R^2	RMSE	MAE
Education-only OLS	0.294	0.294	1,763	1,345
Multivariable OLS	0.545	0.542	1,414	1,050
Ridge	0.545	0.542	1,414	1,050
Elastic Net	0.545	0.542	1,414	1,050
Gradient Boosting	0.603	0.599	1,321	950
Random Forest / XGB	**0.645**	**0.649**	**1,249**	**880**

Why does ML outperform OLS here? In a single-year cross-section, most income variation is explained by a handful of additive factors (education, age, gender), leaving little room for nonlinear gains. The KoWePS panel spans multiple waves and captures household-level dynamics: family size, location, and life-cycle stages interact in ways OLS cannot model additively. Tree-based methods detect these interactions implicitly. This is the key structural difference from single-year cross-sectional analyses where ML rarely beats a well-specified Mincer equation.



(a) Feature group importances. Household size ranks first, not education, revealing the nonlinear household-income channel that powers the ML gain.



(b) Actual vs predicted on the holdout set. Points cluster tightly around the 45° line; residual fan-out at high incomes is expected.

5 Summary

Indicator	Value
Raw dataset	92,857 person-years
Modelling sample	86,313
Spearman ρ (education-income)	0.609
OLS education premium	12.47% per step
Bootstrap 95% CI	[12.0%, 13.0%]
Premium at Q75 vs Q25	14.2% vs 11.5%
Best model holdout R^2	0.645 (Random Forest / XGB)
Best model CV R^2	0.649
Top region	Seoul

H1 confirmed — education carries a large, robust premium after all controls. **H2 confirmed** — the premium rises through the income distribution, implying sorting. **H3 confirmed** — ML wins because the panel contains nonlinear household interactions.

Repository

- `income_welfare_modeling.ipynb` — narrative walkthrough
- `analysis_final.py` — regenerates all 14 figures + 13 tables
- `outputs/figures/` — 14 publication-ready PNG charts
- `outputs/tables/` — 13 CSV outputs incl. bootstrap results
- `docs/korea-income-welfare-brief.pdf` — this document